MSCI 436 • INTELLIGENT DECISION SUPPORT SYSTEMS

Wealth Lens

An Interactive Decision Support System for Portfolio Optimization & Stock Selection

Team 14

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Demo | 2026-08-04 | github.com/s2gouris/wealth-lens

Helping Beginner Investors Make Informed Decisions

1

Who

Beginner & individual retail investors (e.g., students) managing their own investment portfolios.

2

Decision

Which stocks to buy, hold, or sell each period, and how to allocate capital across a portfolio.

3

Consequence

A wrong call means realized losses, missed gains, or a portfolio that is over-concentrated in risk the investor never intended to take.

Scope

Portfolio Selection and Allocation.

This IDSS is an interactive support tool that helps self-directed and beginner investors choose stocks and build portfolios aligned with their risk tolerance and investment horizon.

Investment Decisions with an IDSS

Investment decisions depend on information that is constantly changing and user preferences that change by scenario, making static reports insufficient.

1 Continuously Changing Market Data

- Prices, volume and sentiment update continuously through market hours.
- Economic indicators and company performance are updated over time.
- Models require regular retaining to prevent outdated recommendations.

2 Interactive Decision Making

- Every investor has a different risk tolerance and time horizon
- Users adjust sliders (risk, horizon, diversification cap) and see the portfolio recompute live
- A batch pipeline can't personalize to a scenario the user picks at runtime

Bottom line: a scheduled script and an emailed report can't recompute a portfolio the instant a user changes their risk tolerance or horizon — that live recomputation is what makes this an IDSS, not a pipeline

Four data families, refreshed continuously

T

Technical

Price, volume, rolling returns, SMA/EMA, Bollinger bands, volume vs. moving average

F

Fundamental

P/E ratio, revenue growth, EPS, debt-to-equity, profit margin

M

Macro

Interest rates, inflation/CPI, sector index level & return

S

Sentiment

Aggregated news & social sentiment over a trailing window

Where it comes from

Yahoo Finance API • Alpha Vantage API • Google Finance • Nasdaq database • company quarterly reports • news APIs • Reddit / social posts

Collection strategy

Initial train set: multi-year historical prices, fundamentals and sentiment across the stock universe

Continuous update: daily pulls for price/volume/sentiment; quarterly for fundamentals

Holdout: a rolling set of production entities withheld to validate before deployment

Preprocessing pipeline and honest limitations

Preprocessing

1. Engineer technical indicators (rolling returns, SMA/EMA, Bollinger band position) from raw price/volume series
2. Aggregate news & social sentiment over a trailing window into a single score
3. Normalize features on very different scales (raw price vs. ratios like P/E)
4. Align fundamentals (quarterly) with daily technical/sentiment features via forward-fill

Limitations we're honest about

- Survivorship bias: historical universe skews toward stocks that stayed listed
- Sentiment is noisy and can lag the actual price-moving event
- Fundamentals report on a quarterly lag, so they're stale relative to daily price action
- API rate limits cap how much of the universe can be refreshed intraday
- Historical data under-represents rare regime shocks (e.g., a COVID-style shock)

04 MODEL

Two prediction models feed one optimizer

1

Market Data

Daily price, volume, fundamentals, macro, sentiment

2

Feature Pipeline

Rolling returns, SMA/EMA, Bollinger bands, aggregated sentiment

3

Regressor

Predicts expected % return per stock

4

Classifier

Predicts Buy / Hold / Sell

5

Portfolio Optimizer

Mean-variance post-processing turns predictions into allocations, applying risk & diversification constraints

Why two separate models?

The regressor answers “how much upside?” while the classifier answers “what action?”: buy/hold/sell, plus a binary growth-probability variant. These are different tasks with different error costs, so decoupling them lets each be evaluated and retrained on its own schedule. Both run on 21 features (technical and macro today, plus fundamentals and sentiment once backfilled), comparing XGBoost against LightGBM and selecting per-horizon via time-series cross-validation rather than a single split, which understated real performance by regime-specific luck. Neither the tax-optimization constraint nor the investment horizon needs its own model. Both are handled as constraints and features inside the optimizer.

The numbers, and why they hold up

Horizon	Regressor R^2 (mean)	Classifier accuracy (mean)	Growth ROC-AUC (mean)	Random baseline
1 month	-0.38	43.3%	**0.53**	33% / 0.50
3 months	-0.17	44.3%	**0.59**	33% / 0.50
6 months	0.00	55.6%	**0.67**	33% / 0.50

Why these numbers are meaningful

Funds with far more data typically reach 55 to 60 percent directional accuracy. Our 6-month classifier reaches 55.6 percent against a 33 percent random baseline, and our 6-month growth model reaches 0.67 ROC-AUC against a 0.50 baseline: both are meaningful signal by that standard.

Accuracy rising from 1 to 6 months fits known market behavior: short-horizon moves are dominated by market-wide noise that stock-level features cannot explain, so longer horizons carry more signal.

Regressor R^2 near zero is reported honestly, not hidden: return magnitude is harder than direction, and near-zero R^2 is common in short-horizon return literature. It should improve once fundamentals data, still backfilling, is included; the pipeline picks that up automatically.

+ Advantages

- Optimizer applies risk tolerance and diversification caps at query time; no retraining needed to personalize
- Each model retrains on its own cadence based on observed drift

− Disadvantages

- Two-model pipeline adds latency and maintenance overhead versus a single model
- Buy/Hold/Sell classes are imbalanced, most stock-days are Hold, which needs careful handling

Load-bearing controls, not cosmetic filters

What the user controls

- Risk tolerance (conservative ⟶ aggressive)
- Time horizon (1 month ⟶ 3 years)
- Max position size / diversification cap
- Account type for tax optimization
- Market scenario (bull / bear / recession / high inflation)



Every chart answers a decision, not just a data question

1	**Allocation chart**	Shows the exact percentages to buy/hold/sell right now — the user acts on it directly.
2	**Return with confidence interval**	Reveals the risk band behind the number, so the user sees whether the upside is worth the uncertainty.
3	**Growth vs. benchmark**	Lets the user judge, period over period, whether to keep trusting the recommendation or intervene.
4	**Risk dashboard (Sharpe, drawdown, volatility)**	Flags when a recommended portfolio has drifted outside the user's stated risk tolerance.

IMPACT & FAIRNESS

Backtested against real benchmarks, constrained to be fair

Impact simulation

Benchmarks: S&P 500, TSX Composite, buy-and-hold strategy

Test data: historical out-of-sample stock data, simulated portfolios, backtests from prior years

False positive cost: recommending an underperforming stock → realized portfolio loss

False negative cost: missing a profitable opportunity → opportunity cost

Fairness constraint

Recommendations must stay consistent regardless of a user's income level, age, or investment experience.

The system may only differentiate recommendations using the user's stated financial objectives and risk preferences — nothing else.

Deployment feasibility: models can be deployed today — evaluated on the historical holdout above before every production rollout.

06 OPERATIONALIZATION

From notebook to production

1 Pipeline continuity

Daily API pulls (price/volume/sentiment) and quarterly fundamental refreshes continue automatically after launch — the feature pipeline never stops running.

2 Retraining cadence

Regressor & classifier retrain weekly-to-monthly overnight, triggered when live performance metrics (RMSE, F1) drift past a threshold.

3 User access

Web dashboard front-end, sized to support thousands of concurrent users, with on-demand analysis requests served in under 5 seconds.

4 Infrastructure

Cloud-hosted (e.g., AWS): scheduled batch jobs for ingestion & retraining, plus a low-latency serving layer for real-time dashboard requests.

MONITORING & SUMMARY

Closing the loop after deployment

Model metrics

Regression: RMSE, MAE, R^2

Classification: Accuracy, Precision, Recall, F1

User impact

Recommendation acceptance rate

User engagement

Portfolio growth vs. benchmark

Business

Active users

Retention rate

User satisfaction scores

PortfolioIQ turns a daily flood of market data into a live, personalized buy/hold/sell and allocation recommendation — with the interactivity and continuous retraining a static report could never provide.

Sources & tools

- IDSS worksheet template: Louis Dorard, Ph.D., licensed CC BY-SA 4.0, ownml.co. UI/DSS component added by Sirisha Rambhatla, May 2022.
- Data: Yahoo Finance API, Alpha Vantage API, Google Finance, Nasdaq database, company quarterly reports, financial news APIs, Reddit/social media.
- Benchmarks: S&P 500, TSX Composite Index.
- Project GitHub repository: https://github.com/s2gouris/wealth-lens